Issues in Information Systems
Volume 25, Issue 1, pp. 1-14, 2024

DOI: https://doi.org/10.48009/1_iis_2024_101

Correlating drug overdoses with dark web market activity

Stephen Mancini, *Robert Morris University, ManciniS@rmu.edu*
Emily Kerry, *National Cyber-Forensics and Training Alliance, Ekerry@NCFTA.ne t*
Krista Burnstein, *Internal Revenue Service, kristab1199@gmail.com*
Madison Maynard, *Robert Morris University, mkmst992@mail.rmu.edu*

Abstract

The following addresses the possible correlation between online illicit drug sales and overdoses. The research entailed utilizing historical scraped data taken from five darknet forums and marketplaces which were active from the period of 2015 until 2017. The scraped data was parsed in order to be searched more methodically. From the data, researchers were able to search for specific criteria most likely to be associated with the proliferation of illegal narcotics. The researchers pulled data from the most prolific vendors as well as the most active members on underground forums whose conversations focused on drug sales. As can be seen in the research, while underground forums have been taken down through the action of law enforcement, criminals continue to employ technology in order to profit from illegal narcotic sales. Most importantly, due to global supply and the ability to deliver anywhere, coupled with the use of non-attributable virtual currencies such as Bitcoin, this problem will likely only exacerbate due to the limited resources available to target global enterprises.

Keywords: dark web, online forums, online marketplaces, overdoses

Introduction

The following paper addresses the possible correlation between online drug sales and overdoses across the world. The problem of tracking the buying and selling of illegal narcotics online is fairly recent. The first online market to take advantage of anonymity of both web hosting and online currency was Silk Road in 2011 (Gayathri, 2011). Silk Road was able to take advantage of 'The Onion Routing' (TOR) service which allowed relative anonymity (Gayathri, 2011). The U.S. Navy developed TOR specifically for anonymity therefore one could argue it is a secure gateway to accessing remote resources anonymously (Kaur & Randhawa, 2020).

Coupled with virtual currency, primarily Bitcoin, Silk Road allowed users from anywhere in the world to purchase illegal narcotics and have them shipped anywhere (FBI, 2015). The growth of the Silk Road market was staggering and according to the FBI, "as of September 23, 2013, Silk Road had nearly 13,000 listings for controlled substances, listed under such categories as "Cannabis," "Dissociatives," "Ecstasy," "Intoxicants," "Opioids," "Precursors," "Prescription," "Psychedelics," and "Stimulants." (FBI, 2015)". While Ross Ulbricht was arrested and found guilty of his role in running Silk Road (FBI, 2015), this effort has spawned more sites and even better lessons learned for criminal actors. As a result, it is unclear how many active marketplaces are currently operating, and it is more unclear how many individuals are using the internet to purchase both legal and illegal narcotics. In a recent study it was stated, "The online

pharmacy market has undergone significant growth during the past two decades as society is shifting to a more digitalized world" (Fittler, Adeniye, Katz, & Bella, 2021, pg. 1).

In addition, while the primary focus of this study was on darknet marketplaces, other researchers noted how little research there was regarding the proliferation of online drugs on more common social media platforms (Demant, Bakken, & Hall, 2020). One can argue that these public sites are overlooked due to the fact darknet sites are often specifically designed for illicit sales and there is an assumption social media sites police themselves. However, because of the breadth of this problem, this paper only leveraged historical underground marketplace data to correlate the impact of this more well-known method of purchasing illegal narcotics.

Background of the Problem

With the advent of the dark web and the ability to conduct online sales with relative anonymity, there has been an increase in drug sales and as a result, probable overdose deaths. For example, in 2016, 63,362 overdose deaths involved an opioid (Scholl L, Seth P, Kariisa M, Wilson N, & Baldwin G, 2018). The drug that was most often used in overdose deaths was methadone (Methadone, 2022; Scholl et al., 2018). Methadone is also known as Fizzies with MDMA, Amidone, Chocolate Chip Cookies, and Wafer (Methadone, 2022). As a highly addictive drug, methadone can be easily abused and misused.

Another drug that gained a reputation as one of the most dangerous drugs is fentanyl. Fentanyl is used medically to manage pain and increase the quality of life for patients for whom it is prescribed (Hinckley, J. D., 2023). While properly manufactured and prescribed fentanyl decreases pain, fentanyl that is manufactured illegally is much different. The fentanyl made illegally is thought to be "50-100 times [more potent] than that of heroin and morphine, respectively" (Hinckley, J. D., 2023). As a result of the high that adolescents and adults get when mixing fentanyl with other drugs, it becomes heavily abused. Between 2011 and 2016, fentanyl was one of the top 10 most frequently used drugs in overdose cases (Hedegaard et al. 2018). AlphaBay, a darknet marketplace operating from 2014 to 2017 amounted to a prominent fentanyl trafficking organization. The marketplace contributed to an early rise in manufacturing counterfeit prescription drugs containing fentanyl (Humphreys, Shover, Andrews, Bohnert, Bandeau, & Caulkins 2022). This rise proved disastrous due to fentanyl's high potency, inducement of respiratory muscle stiffness, and other attributes that increase the risk of overdose (Gill, Kelly, & Henderson 2019).

The fentanyl problem continued to rise within AlphaBay, when a Utah-based organization consisting of Aaron Shamo and Drew Crandall started selling various drugs (Galofaro & Whitehurst, 2019). The group's main source of sales were counterfeit 'Oxycodone' pills (which contained fentanyl but no Oxycodone) and counterfeit Xanax pills, which apparently did contain Alprazolam (Caulkins et al. 2023). By the fall of 2016, the group shipped to customers worldwide, selling roughly 300,000 pills per month. See appendices C and D to gain further perspective on weekly drug sales. One indication of the organization's size is that prosecutors were able to identify 90 of their customers who were deceased as of the time of sentencing, with confirmation that the cause of death was an overdose in some cases (Caulkins et al. 2023).

Another individual by the name of Henry Koffie, has completed more than 7,800 fentanyl sales in less than two years (Central Oregon Daily, 2023). The seller is linked to dozens of fatal overdoses across the United States. In one year, three Portland residents overdosed on furanyl fentanyl supplied by Koffie (Central Oregon Daily, 2023). The first victim was a 19-year-old student at Portland State University who, on May 2, 2016, overdosed 30 minutes after ingesting powdered furanyl fentanyl. Paramedics administered Naloxone and breathing assistance to the student, saving her life (Central Oregon Daily, 2023). A year later, officers responded to a fatal overdose of a 27-year-old in southeast Portland. The victim's roommates

told the officers that the victim had ordered fentanyl from Narcoboss on AlphaBay who had advertised the furanyl fentanyl as "China White" (Central Oregon Daily, 2023). Three weeks later, on May 29, 2017, officers responded to another fatal overdose of a 27-year-old who had resided in northeast Portland. A small vial of furanyl fentanyl had been located in the victim's residence as well as a notebook containing information on accessing AlphaBay and a Bitcoin wallet. (Central Oregon Daily, 2023). By July 2017, the Justice Department, in partnership with the FBI, United States Drug Enforcement Administration (DEA), HSI and IRS-Criminal Investigation (IRS-CI), seized and shut down AlphaBay, which, at the time, was the largest criminal marketplace online (Central Oregon Daily, 2023).

Naloxone was created in the 1970s as a way to reverse the effects of opioid overdose but was only allowed to be used in medical facilities. In March 2016 Canada passed legislation that allowed naloxone to be sold over the counter to the public (Campbell, 2019). In March 2023, the U.S. Food and Drug Association made naloxone available over the counter (White House, 2023). According to Cambell, "150 preventable deaths occur [in the United States] and close to 1000 people are treated for opioid-related injuries" (Campbell, 2019). Naloxone, if it needed to be used in a home, had to be prescribed by a doctor and was only prescribed to those that needed to take heavy opioid drugs for extended periods (U.S. Food and Drug Association, 2023).

In 2010, Massachusetts was the first state within the United States to give naloxone to its first responders. Since then, more states have adopted this practice and first responders typically carry naloxone. Overdose deaths have only increased since 2016 with the most recent data showing that "more than 101,750 reported fatal overdoses occurring in the 12-month period ending in October 2022, primarily driven by synthetic opioids like illicit fentanyl" (U.S. Food and Drug Association, 2023). The number of overdose deaths has been increasing over the years and prevention measures have done little to impact the death toll. The key importance to this information is that clearly, with an increase of overdose, this is implying an increase in availability.

Problem Studied

Online drug sales and overdose deaths are still an unfortunate part of society and must be researched in depth in order to make strides in combating this epidemic (National Institute on Drug Abuse, 2023). With the proliferation and ease of accessibility to illegal narcotics made available via the darknet, the reality is that untested/unknown drugs are likely making their way into all corners of society. In 2023, it was reported that over 112,000 fatal overdoses occurred in the United States with fentanyl being one of the primary drivers (Mann & Pattani, 2023). As the drugs primarily originate from all over the world, the darknet provides access to anyone who has access to the internet. Therefore, the general problem addressed here is the impact of the online proliferation of illegal narcotics and their probable relation to overdoses.

> **RQ**: *How can the online illegal sales of opioids, and other illegal narcotics, be associated to the rise of overdose cases?*

Methodology

The analysis consisted of both qualitative and quantitative aspects. The overall approach we employed consisted of sampling the most active conversations involving drug usage/sales and vendors with the most sales taken from five dark web sites (listed below) coupled with overall sales/overdose statistics during the

same timeframe. This first part required a more subjective/qualitative approach. As drug vendors use various online 'lingo' for their interactions, researchers had to manually review the more active participants and identify any commonalities of verbiage or search for keywords. For example, vendors typically use XTC for ecstasy. From this, we were able to provide more direct queries into the database. In addition to manual review, a thorough study of DreamMarket was conducted which allowed us to immediately focus on the sold narcotics (Gengqian, Jianwei, Yunqian, Du, & Shuqiang, 2020). For example, researchers identified weed, cocaine, hash, and MDMA as the top four most available items (Gengqian et al., 2020). Therefore, our search from all five markets would include, at a minimum, these keywords.

As noted above, vendors typically have certain words which mean certain drug types and they may not say certain drugs names directly. For example, it is almost unheard nowadays for online vendors to explicitly say they are selling fentanyl due to federal interest. In 2023 alone, there were over 112,000 fatal overdoses and with the DEA hosting an Annual Family Summit on Fentanyl at their Headquarters, it is not the attention online vendors are typically looking for (Mann & Pattani, 2023). In addition, vendors will use 'slang' and other words which make direct queries difficult if researchers are not aware of the verbiage. See Figure 1 for an example posting from DreamMarket circa 2016.

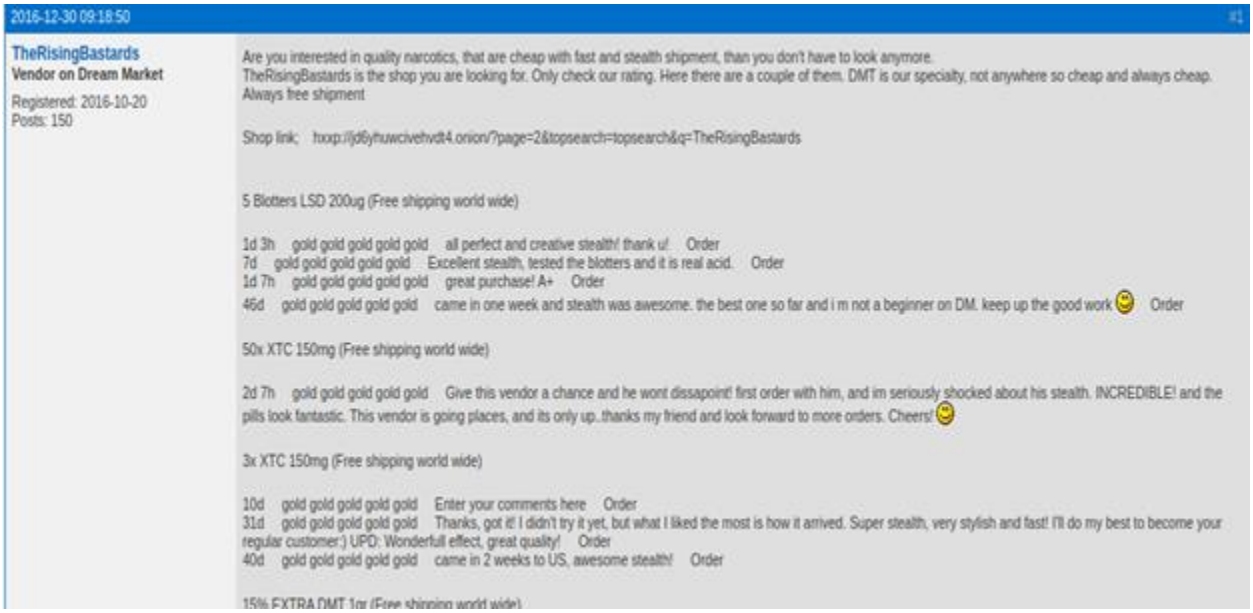


Figure 1: Screenshot of Typical Vendor on Dream Market

The research database utilized was created by parsing the scraped data from five darknet sites retrieved when the sites were live. While the sites are no longer available online due to various law enforcement and other actions, the researchers have saved significant portions of the sites in order to achieve a relevant sampling of data which is significant enough to draw some conclusions. For example, AlphaBay which had been operating since 2014, was taken down by the FBI in 2017 and researchers were able to search well over 1,000 vendor products (FBI, 2017).

Once search words were identified, researchers were able to pinpoint key vendors and other relevant posts. From this, they were able to ascertain the number of sales from various high-level operators and which drugs were most prevalent. While they were unable to directly connect specific sales to specific deaths, they were able to show whether online sales increased and at the same time, were there overdose deaths

being reported. The statistical analysis employed a direct sampling using general trends during the years data from the darknet sites was available.

Data Sources

In 2013 when the Silk Road site was taken down, there were roughly 2.7 billion internet users (Duarte, 2023). In 2017, there were 3.7 billion users (Duarte, 2023). In 2023, that number has grown to 5.2 billion users, a nearly 74.1% increase (Duarte, 2023). At the same time, the number of users on the internet has increased, the number of drug sales and overdoses has also increased (Duarte, 2023; National Institute on Drug Abuse, 2023). The use of online forums and crypto-markets has become one of the easiest ways for drug users and sellers to traffic drugs. With the use of encryption and anonymity, it allows for users to feel confident that they will not be found. Some examples of illegal websites that allowed this were: DreamMarket, DNM Avengers, AlphaBay, Valhalla, etc. It is through the data that we collected from these sites that we could see trends in drug sales and which drugs were more popular.

The timeframes from the scraped websites was between approximately 2015 and 2017 and entailed over 35,000 posts. The sites were parsed using customized Python scripts. Again, Appendix A provides a sample of the output after the raw data is parsed. To ascertain which vendors were considered the most active during said time frame, we queried the resultant data extracted via our custom-built parsing tools. With this data, we were able to establish the number of drug sales made by the top vendors on these sites. Once these key actors were identified, we utilized overdose deaths during the same timeframe and performing basic statistical analysis, we attempted to determine any correlations. Unlike the examples provided above, while it cannot be said definitively whether or not that particular individual is responsible for any of the reported deaths, it can demonstrate how increased online sales are probably responsible for deaths or overdoses. The data for the study consisted of data from the following sites which was scraped during the timeframe they were active on the dark web:

1. AlphaBay
2. DNMAvengers
3. Dream Market
4. Valhalla
5. Hub

Analysis and Discussion

As noted above, the data collection consisted primarily of leveraging previously scraped dark web marketplace and forum data. In addition, the marketplaces allowed access to top vendor sales history and ratings (Figure 2). The process for cleaning up the data consisted of developing custom Python scripts. As noted above, the initial review involved manually reviewing the data to identify either the most active individuals or identifying the most potentially lethal drugs (see appendix B for an example). We break out several top posters and provide a sampling of their comments we deemed were most relevant (appendix B). For example, if you take a look at the individual with the moniker name, ALauizen, they posted 38 times and were selling Diazepam. On May 9, 2016, ALaurizen posted:

> *"I offer products of excellent quality and at very reasonable prices..Super discounts for bulk order$$Please, visit any of our vendor pages , and do not hesitate in asking what you want to know!In stockENERGY CONTROL TESTEDYou can see EC results here Diclazepam (analog of*

*Diazepam) hxxp://**MASKED**.onion/index. I have very extensive experience in E.commerce.Specialized in bulk shipments.Great customer support (24h).Prices are very convenient for resellers.Constant stock of product.We offer tracking for all shipments.Shipping from China.AVERAGE DELIVERY TIMESUSA - 9 / 12 Business DaysEU/UK - 8 / 10 Business Days".*



Figure 2: Small screen shot here of a top vendor from AlphaBay

In addition to being active on these markets, vendors likely operated out of multiple locations. For example, the individual known as *excalibur* had 385 posts on DreamMarket. Some of them clearly indicated the individual was also active or aware of others from older forums which either were taken down by law enforcement or shut down on their own accord. In a post that was dated 2015-11-08 on DreamMarket, *excalibur* stated, "*any refugees from Abraxas or Agora?if so let's talk about heroin vendors and other related nonsense…*" As such, while reviewing these markets provides clear insight into the depth of activity occurring online, there was far more activity occurring. Table 1 provides a numerical breakout sampling of some top vendors in 2015, 2016, and 2017 who sold opioids w/ fentanyl as annotated by the dark web marketplaces.

Table 1 - Top Seller Sampling Break-Out of Fentanyl Sales

Seller	2015	2016 (6 months)	2017	Comment Total*	Market
Pharmaphil	N/A	513	604	N/A	AlphaBay
Pharmaphil	N/A	N/A	N/A	863	Valhalla
QualityKing	N/A	N/A	251	N/A	AlphaBay
Light123	N/A	45	112	N/A	AlphaBay

* Year breakout on Valhalla is unknown but based on scraped dates, total of sales is equated to total customer feedback/Not applicable for AlphaBay

According to the National Institute on Drug Abuse, a branch of the National Institutes of Health (NIH), overdose deaths in 2015, 2016, and 2017 were 52404, 65000, and 70000 respectively (National Institute on Drug Abuse, 2023). Of those overdose deaths roughly, 12112, 20000, and 25000 were due to stimulants like cocaine (National Institute on Drug Abuse, 2023). *FrankMatthews*, a seller on AlphaBay had made over 5000 sales from 2016-2017. *FrankMatthews* sales ranged from high quality cocaine to different kinds of ecstasy. He was busy on AlphaBay and was trusted by most of the community, having a vendor and trust level of 10 and an overall positive feedback rating at 95%.

In a more recent study from 2021 the Center for Disease Control and Prevention (CDC) found that "synthetic opioid-involved death rates increased by over 22% from 2020 to 2021 and accounted for nearly 88% of all opioid-involved deaths in 2021 (*Opioid Overdose | Drug Overdose*, 2023.)". This would suggest that even with the current measures in place to try and decrease the opioid epidemic it has been futile. As such, another major vendor on AlphaBay was *MajorKey*, who had made over 2000 sales from 2016-2017. Comments for this vendor were overwhelmingly positive and they scored a 98% on positive feedback. This vendor has a foothold in the cocaine market and was a known seller but then took a break in order to "regroup and optimize."

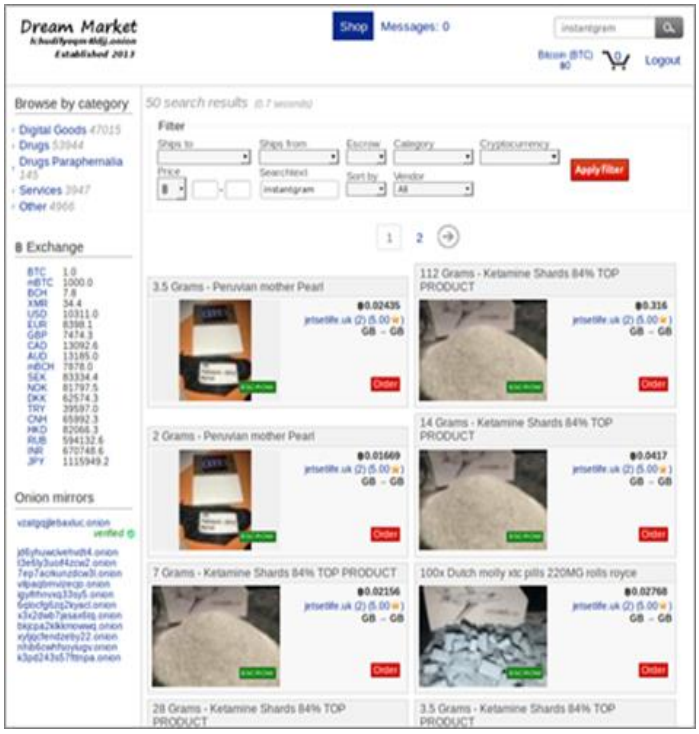


Figure 3: Small screen shot here of a top vendor from DreamMarket

General Findings

Analysis of the data showed that there was a significant amount of drug sales which occurred between these timeframes from the sites which were sampled. While there exists 'scammers' on these sites, it is clear by the ratings and reviews that there were a large number of legitimate sellers provided various illegal narcotics to individuals all over the world, from all over the world. Therefore, while drug sales, drug overdoses, and

internet users have a positive correlation to each other but this does not mean that they have a direct link with each other. With over 33810 collective posts from five different websites, it is clear that these forums were active and contributed to the distribution of drugs worldwide. The themes are all indicative that drug forums are used as a way to sell, buy, and review opioids, hallucinogens, and other products. As a result of the increase of users on the internet, there is a higher chance that there has been an increase in the number of users on the dark web. Therefore, there are more sales and a more likely chance that new users may overdose, or old users will have a new way to get the drugs they require in a non-traceable, anonymous forum.

Conclusion

The study addressed the correlation between drug overdoses and activity on drug forums. While the study only leveraged a relatively small sampling of previously active sites to demonstrate the impact of only a few sellers. As drug forums were used more drug overdoses also increased. This could be attributed to the increase in the number of users on the internet or an increase in the number of drug users. While the study is not conclusive as the data sampling was minor compared to actual numbers of online drug sales, between identified legal cases and the sheer volume of sales, there is no doubt that the online sale of illegal narcotics is contributing to overdoses.

Future Research

The purpose of this study was to identify any potential correlation between online drug sales and overdoses. Because it was limited to online sales, this research did not attempt to address the impact of non-online sales with respect to overdose deaths. As was clearly noted, the growth of online sales continues to climb as does overdose deaths in America (Mann & Pattani, 2023). Therefore, several key areas of research are needed in order to attempt to stem the real impact of online drug sales. First, this dataset, while fairly robust, only entailed five websites. Further studies are needed to ascertain if other sites are more directly associated with sales to overdoses, likely due to the lethality of the drugs being sold. In addition, once the sites are taken offline through law enforcement action, identifying the most prolific buyers to determine if they are merely middle-men or users would better determine if online drug markets have simply replaced manual movement of drugs through mules. For example, with the employment of postal services for shipping, while risky, does this reduce the potential cost of hiring personnel to move narcotics to the street level? Again, further studies on this would be warranted. Finally, online vendors are beginning to take advantage of more peer-to-peer transactions thus reducing information available to law enforcement. So, peer-to-peer would not allow researchers or law enforcement to identify top vendors thus leading to placing resources on lesser targets. As such, is there a way to identify the impact of overdose deaths via peer-to-peer transactions and can individuals be identified via other means?

References

Broséus, J., Rhumorbarbe, D., Mireault, C., Ouellette, V., Crispino, F., & Décary-Hétu, D. (2016). Studying illicit drug trafficking on Darknet markets: structure and organisation from a Canadian perspective. *Forensic science international*, *264*, 7-14. Retrieved 12 27, 2023, from https://doi.org/10.1016/j.forsciint.2016.02.045.

Campbell, N. D. (2019). Naloxone as a technology of solidarity: history of opioid overdose prevention. *CMAJ*, *191*(34), E945-E946.Retrieved December 18, 2023, from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6710079/

Campbell, R., Goodman-Williams, R., Feeney, H., & Fehler-Cabral, G. (2020). *Assessing triangulation across methodologies, methods, and stakeholder groups: The joys, woes, and politics of interpreting convergent and divergent data*. The American Journal of Evaluation, 41(1), 125–144. https://doi.org/10.1177/1098214018804195

Caulkins, Jonathan P., Schicker, Philippe C., Milward, H. Brinton & Reuter, Peter (2023) A detailed study of a prominent dark web fentanyl trafficking organization, Global Crime, DOI: 10.1080/17440572.2023.2291352

Central Oregon Daily News Source. (2023, December 11). DOJ: Life sentence for man dealing fentanyl analogue that caused 2 Oregon deaths. Retrieved January 9, 2023 from https://centraloregondaily.com/fentanyl-analogue-life-sentence/#:~:text=%E2%80%94A%20Pennsylvania%20man%20who%20completed,AlphaBay%2C%20a%20former%20darknet%20marketplace%2C

Demant, J., Bakken, S. A., & Hall, A. (2020). Social media markets for prescription drugs: platforms as virtual mortars for drug types and dealers. [Social media markets for prescription drugs] *Drugs and Alcohol Today*, *20*(1), 36-49. https://doi.org/10.1108/DAT-06-2019-0026

Duarte, F. (2023, November 22). *Countries with the Highest Number of Internet Users (2024)*. Retrieved December 18, 2023, from https://explodingtopics.com/blog/countries-internet-users#time

FBI. (2015, February 2). *Ross Ulbricht, the Creator and Owner of the Silk Road Website, Found Guilty in Manhattan Federal Court on All Counts*. https://www.fbi.gov/contact-us/field-offices/newyork/news/press-releases/ross-ulbricht-the-creator-and-owner-of-the-silk-road-website-found-guilty-in-manhattan-federal-court-on-all-counts

FBI. (2017, July 20). *Darknet Takedown: Authorities Shutter Online Criminal Market AlphaBay* https://www.fbi.gov/news/stories/alphabay-takedown . Retrieved December 23, 2023.

Fittler, A., Adeniye, L., Katz, Z., & Bella, R. (2021). Effect of Infodemic Regarding the Illegal Sale of Medications on the Internet: Evaluation of Demand and Online Availability of Ivermectin during the COVID-19 Pandemic. *International journal of environmental research and public health*, *18*(14), 7475. https://doi.org/10.3390/ijerph18147475

Gill, H., Kelly, E., & Henderson, G. (2019). How the complex pharmacology of the fentanyls contributes to their lethality. *Addiction*, 114(9), 1524. https://doi.org/10.1111/add.14614

Hedegaard, Holly et al. (2018). Drugs most frequently involved in drug overdose deaths: United States, 2011-2016

Humphreys, K., Shover, C. L., Andrews, C. M., Bohnert, A. S., Brandeau, M. L., Caulkins, J. P., Chen, J. H., Cuéllar, M. F., Hurd, Y. L., Juurlink, D. N., Koh, H. K., Krebs, E. E., Lembke, A., Mackey, S. C., Larrimore Ouellette, L., Suffoletto, B., & Timko, C. (2022). Responding to the opioid crisis in North America and beyond: Recommendations of the Stanford–lancet commission. *The Lancet*, 399(10324), 555–604. https://doi.org/10.1016/S0140-6736(21)02252-2

Gayathri, A (2011, June 11). *From marijuana to LSD, now illegal drugs are delivered on your doorstep*. https://www.ibtimes.com/marijuana-lsd-now-illegal-drugs-delivered-your-doorstep-290021

Gengqian, Z., Jianwei, Z., Yunqian, F., Du, K., & Shuqiang, L. (2020). A Market in Dream: the Rapid Development of Anonymous Cybercrime. *Mobile Networks and Applications, 25*(1), 259-270. https://doi.org/10.1007/s11036-019-01440-2

Hinckley, J. D. (2023). *Fentanyl: Accelerant of the Adolescent Opioid Crisis*. Psychiatric Times, 40(6), 24–25.

Kaur, S., & Randhawa, S. (2020, January 28). *Dark Web: A Web of Crimes. Springer Lin*k. Retrieved December 23, 2023, from https://link.springer.com/article/10.1007/s11277-020-07143-2

Mann, B., & Pattani, A. (2023, December 28). *In 2023 fentanyl overdoses ravaged the U.S. and fueled a new culture war fight*. https://www.npr.org/2023/12/28/1220881380/overdose-fentanyl-drugs-addiction

Methadone. (2022, October). DEA.gov. Retrieved December 18, 2023, from https://www.dea.gov/factsheets/methadone

National Institute on Drug Abuse. (2023, July 30). *Drug Overdose Death Rates*. NIDA.NIH.GOV | National Institute on Drug Abuse (NIDA). Retrieved December 28, 2023, from https://nida.nih.gov/research-topics/trends-statistics/overdose-death-rates

Scholl L, Seth P, Kariisa M, Wilson N, & Baldwin G. (2018). *Drug and Opioid-Involved Overdose Deaths - United States, 2013-2017*. MMWR Morb Mortal Wkly Rep. 2018 Jan 4;67(5152):1419-1427. doi: 10.15585/mmwr.mm675152e1

U.S. Food and Drug Association. (2023, March 29). *FDA Approves First Over-the-Counter Naloxone Nasal Spray*. Retrieved 12 18, 2023, from https://www.fda.gov/news-events/press-announcements/fda-approves-first-over-counter-naloxone-nasal-spray

White House. (2023, March). Dr. Gupta Applauds FDA Announcement to Make First Naloxone Product Available Over-the-Counter. Retrieved Jan 2023 from https://www.whitehouse.gov/ondcp/briefing-room/2023/03/29/dr-gupta-applauds-fda-announcement-to-make-first-naloxone-product-available-over-the-counter /

Appendix A - Sample of Scraped and Parsed Forum Data

Date	Moniker	Posts	Comment
2016-12-19 06:29:33	TheRisingBastards	Posts: 150	Just check our ratings!!!Fast,free and stealth shipping worl...
2016-12-27 02:44:11	TheRisingBastards	Posts: 150	Just check our ratings!!!Fast,free and stealth shipping worl...
2016-12-28 04:56:53	TheRisingBastards	Posts: 150	FREE SHIPMENT WORLDWIDE
2016-12-11 22:26:14	Marketplacerr	Posts: 1	1.AlphaBay Marketplace = hxxp://-scrubbed-/affiliate.php_r...
2015-03-27 08:17:30	whosyours	Posts: 50	hxxp://-scrubbed-/ - Darknet bitmixer.iohxxp://-scrubbed-/ - ...
2015-04-05 16:17:09	benjamindover	Posts: 18	I have always heard your should only access bit mixer sites ...
2015-04-06 10:59:55	whosyours	Posts: 50	benjamindover wrote:I have always heard your should only ...
2015-04-06 15:25:36	kitt3n	Posts: 817	benjamindover wrote:I have always heard your should only ...
2015-04-16 08:30:16	whosyours	Posts: 50	yes, use TOR for sure
2015-04-16 14:03:30	Parsonala	Posts: 1	Use Tor + VPN.When using Tor, i recommend using VPN.Re...
2015-04-20 04:21:35	SpeedStepper	Posts: 443	Hi there,i give a warning on Helix, i got an incident that mon...

Appendix B - Sample Moniker Review Outcome

Moniker	Posts	Site	Sample Post / Notes
GregoryHouse**	3802	Hub	**The closest person on Hub was Dodongo who appeared to be a moderator/seller who posted 3254 times.
ShroomPrince	296	Hub	2017-01-04: *I know that I am a vendor here but I see this forum is lacking a mushroom review thread. I'm hoping that this will start one. Leave your reviews thoughts suggestion for me or any other DN mushroom vendor.*
ALaurizen	38	DreamMarket	2016-05-11: *Yes, I have bulk stock ofMT-45U-47700Furanyl-Fentanyl (fentanyl analogue)Please, visit any of our vendor pages , and do not hesitate in asking what you want to know!AL*
Etherreality	1125	DreamMarket	*I was just informed by Luis at EnergyControl that heroin containing possible fentanyl or fentanyl derivatives will be tested at no charge to the user or the Avengers[...]If you suspect an order that you made contains heroin, please email international /at/ energycontrol.0rg.[...] They will only tell you whether fent is present or not. If you want analysis done on your heroin, message a Global Moderator for a code.Ethereality*
PachoHerrera	285	DreamMarket	*Also Bulk listings MDMA online150G - 850 euro200G - 1125 euro250G - 1400 euro500G - 2500 euro1000G - 4500 euroWe can give tracking for every country but not for USA, AUS, NZ, ISREALFor further questions just send us a PM!Or for great custom deals!RegardsPachoHerrera*

Moniker	Posts	Site	Sample Post / Notes
KetaMaster	25	Hub	2016-12-24*: Hi there mates !I would like to introduce me :I work from many years with the ketamine and this is the best quality you can ever get . Is 97% purity shining long crystals, just have a look at the full quality photos: You can see that this is totally different from other Ketamine (Dutch, Indian or African)We have the test results from a laboratory that is 97% . We offer also free samples to people that are interested in it.97% KETAMINE LAB TEST:We are very glad to offer a top quality product at a good price.*

Appendix C

Weekly sales volume by drug type, in dollars. (Caulkins et al. 2023)

